$\Lambda \to p\pi^-$ in the ePIC B0 tracker: update

Hit seeding (EICrecon#3116), the 5 keV default, and Λs up to 275 GeV

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Code, results, reports: `https://github.com/gadininio/eic-B0-detector`
Previous slides: `docs/slides/slides.pdf` (3 November 2026)

Reminder and what changed since 3 November

Reminder: single-Λ gun, $\theta = 8$–18 mrad; $\Lambda \to p\pi^-$ reconstructed in the B0 tracker; old B0 (epic `43f427e`, 300 μm) vs realistic B0 of eic/epic#1194 (`ec774d8`, 50 μm); truth-seeded tracks, EICrecon main `8b44cae`, 50–100 GeV.

	3 November	now
B0 seeding	truth (#2901)	truth **and hit seeding (#3116)**, same job, same events
EICrecon	main `8b44cae`	#3116 branch `1ca147b` (main of 11-11 + #3116)
B0 threshold	10 keV default, 5 keV by option	5 keV default (EICrecon#3084)
eic-shell	nightly 11-02	nightly 11-07 (new B0 in its epic)
old geometry	from the container	local build of the same commit
p_Λ	50–100 GeV	50–100 (same events) + **100–275 GeV**

- #3028 (first B0 seeding PR) closed 11-05 in favour of #3116 (open); #3029 was a bot fix into the #3028 branch, not main

Hit seeding (EICrecon#3116)

- ACTS triplet seeder: one hit in each of **3 B0 stations**
- Hits rotated so the dipole field at the middle hit is along z
- No vertex or charge assumption $\Rightarrow$ works for displaced decays; min. p across the field 1 GeV, ≥ 3 measurements
- `B0TrackerSeeds` $\rightarrow$ `B0TrackerCKFTracks` $\rightarrow$ `CombinedTracks` $\rightarrow$ `ReconstructedChargedParticles`
- Truth-seeded chain runs in parallel: `ReconstructedTruthSeededChargedParticles`
- Seed $\leftrightarrow$ daughter: ≥ 2 of the 3 seed hits made by the daughter
- Macros: new argument `seeding = "truth" | "hit"`

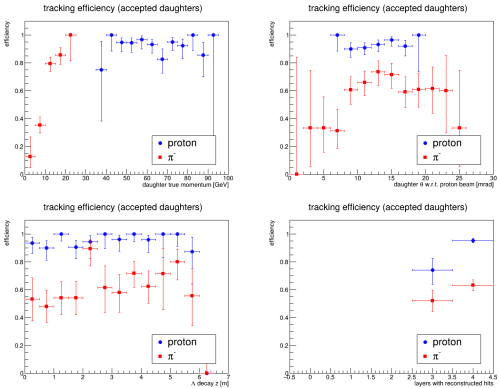
Tracking efficiency: truth vs hit seeding (50–100 GeV)

accepted daughters, p / π^-	old, truth	old, hit	new, truth	new, hit
accepted	404 / 349	404 / 349	295 / 240	295 / 240
seed	99 / 99 %	100 / 100 %	97 / 96 %	100 / 100 %
CKF track	98 / 66 %	100 / 74 %	94 / 61 %	100 / 67 %
reconstructed particle	**97.5 / 66.5 %**	**99.5 / 72.5 %**	**93.2 / 60.8 %**	**100 / 64.6 %**

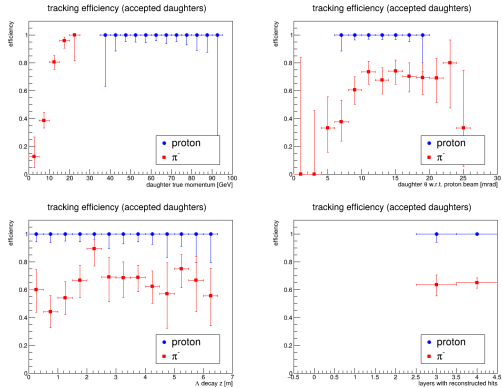
- Hit seeding: a seed for **every** accepted daughter, **every** accepted proton reconstructed
- The π^- is still lost in the **CKF**: 5–10 GeV π^- 39 % (new), 48 % (old) with hit seeding
  $\Rightarrow$ **the soft-π^- loss is not a seeding problem**

Tracking efficiency, new geometry

truth seeding



hit seeding



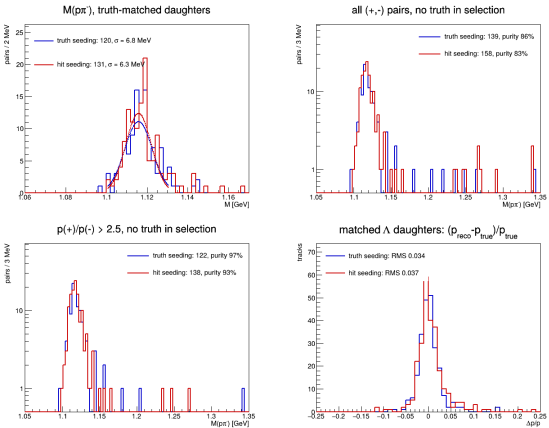
vs daughter momentum, θ, decay z, layers with rec. hits; proton blue, π^- red

Acceptance × efficiency: 3 November vs now

	3 Nov: main, truth	#3116 build, truth	#3116 build, **hit**
old geometry			
tracking eff. p / π^-	82.2 / 55.6 %	97.5 / 66.5 %	99.5 / 72.5 %
$A \times \varepsilon$ both / $p\pi^-$ decay	29.8 %	41.7 %	45.5 %
both rec. / generated Λ	15.0 %	21.0 %	**22.9 %**
new geometry (3 Nov: 5 keV by option)			
tracking eff. p / π^-	92.9 / 60.8 %	93.2 / 60.8 %	100 / 64.6 %
$A \times \varepsilon$ both / $p\pi^-$ decay	24.7 %	24.7 %	27.0 %
both rec. / generated Λ	12.0 %	12.0 %	**13.1 %**

- New geometry: truth seeding reproduces 3 November exactly; hit seeding +1.1 points
- Old geometry, not understood: same simulation, truth seeding, efficiency jumps with the new build. Differences: EICrecon `8b44cae` → `1ca147b`, eic-shell image (ACTS), local epic build. Not the threshold

Λ mass: truth vs hit seeding (same events)



new geometry; blue truth, red hit seeding

	old	new
pairs, truth / hit	210 / 229	120 / 131
σ [MeV], truth / hit	6.9 / 6.8	6.8 / 6.3
$\Delta p/p$ RMS	3.8 / 4.4 %	3.4 / 3.7 %
purity after cut	86 / 64 %	97 / 93 %
$\pm$20 MeV: sig./cand.	206 / 219	116 / 116

- Same peak: M(hit) − M(truth) RMS 4.4 MeV on common decays
- Hit seeding also tracks secondaries: "other" tracks old 198 → 432, new 68 → 91
- Background matters only in the thick old geometry

Correction: per $p\pi^-$ decay vs per generated Λ

1000 Λ, 50–100 GeV, new geometry:

fate	N	$z_{\mathrm{end}} < 5.8$ m	last z_{end}
$p\pi^-$	485	438	8.7 m
$n\pi^0$	278	246	9.4 m
other	237	11	12.4 m

- "other" = **interacted** (all have daughters), 205 at $z = 6$–10 m; none leaves the world (3 Nov: wrong)
- A Λ not decayed by $\sim$6 m interacts before it can decay
- "90 % of $p\pi^-$ decays before B0" is **conditional**
- Per generated Λ: 68 % decay before 5.8 m vs **66 %** from $1 - e^{-L/\lambda}$

$\Rightarrow$ **per generated** Λ is the unbiased denominator

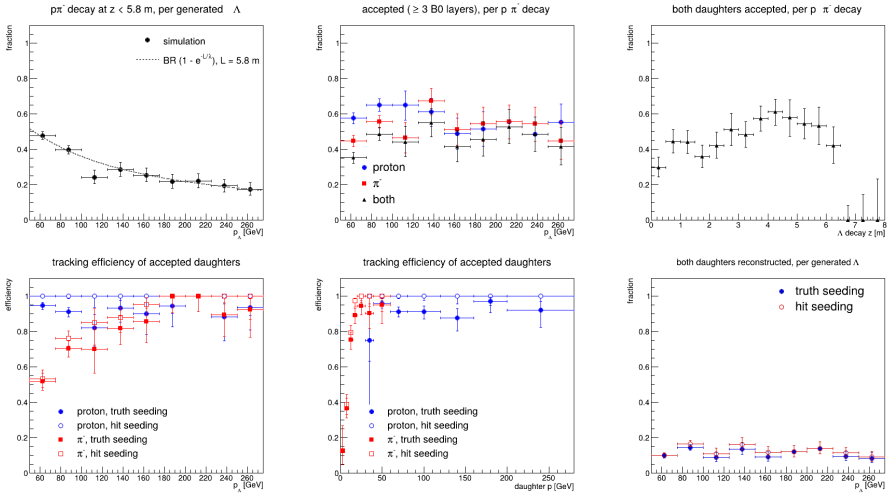
Λs up to 275 GeV: expectation

- New samples: 100–275 GeV, 2×500 per geometry, #3116 build (both seedings)
- Mean decay length $\lambda = (p/m)\, c\tau = p[\text{GeV}] \times 7.07\,\text{cm}$
- Decays before the B0 ($L = 5.8\,\text{m}$): $P = 1 - e^{-L/\lambda}$

p_Λ	50 GeV	100 GeV	200 GeV	275 GeV
λ	3.5 m	7.1 m	14 m	19 m
decay before B0	81 %	56 %	34 %	26 %

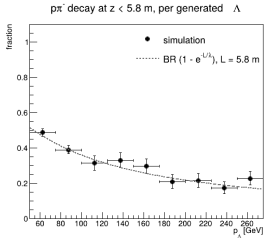
- But the π^- carries 6–25 % of p_Λ $\Rightarrow$ no longer soft at high momentum

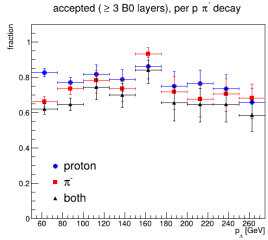
Acceptance and efficiency vs Λ momentum (new geometry)

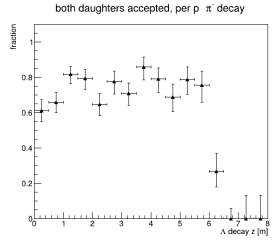


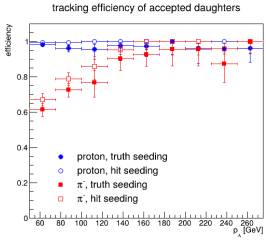
top: $p\pi^-$ decay before B0 (dashed: decay law), acceptance vs p_Λ, vs decay z; bottom: tracking eff. vs p_Λ and vs daughter p (filled truth, open hit), both reconstructed per generated Λ

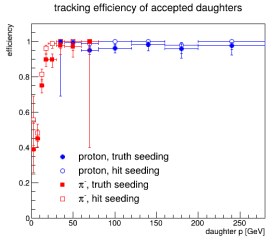
Acceptance and efficiency vs Λ momentum (old geometry)

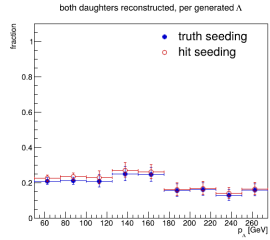












Energetic Λs: roughly flat

p_Λ [GeV]	$p\pi^-$ decay before B0 (decay law)	π^- eff., truth / hit		**both rec. / generated** Λ, truth / hit	
		old	new	old	new
50–75	48 % (47)	62 / 67 %	52 / 53 %	21 / 23 %	10 / 10 %
150–175	25–30 % (25)	93 / 100 %	86 / 95 %	25 / 26 %	9 / 12 %
250–275	18–23 % (17)	100 / 100 %	93 / 100 %	16 / 17 %	9 / 10 %

- Decay before the B0 follows the decay law
- π^- efficiency vs its own momentum: ≈40 % below 5 GeV → ≈100 % above 20 GeV
- Acceptance (both, per decay): old 60–85 %, new 35–55 % (ϕ gap, not momentum); nothing beyond $z \approx 6.5$ m
- **Fewer early decays, but better reconstructed:** new geometry 21 % of early decays fully reconstructed at 50–75 GeV, 40–60 % above 150 GeV
- 30–55 $p\pi^-$ decays per bin above 100 GeV: ±5–10 points

Summary

- **Hit seeding (EICrecon#3116) works** for displaced Λ daughters: seed for every accepted daughter, every accepted proton reconstructed; 13.1 % (new) / 22.9 % (old) of all Λ fully reconstructed (truth seeding 12.0 / 21.0 %); same mass peak
- It also tracks secondaries: purity without truth 93 % (new), 64 % (old, thick sensors)
- **Soft-π^- loss is in the CKF**, not in the seeding
- **5 keV** is now the default B0 threshold (#3084)
- **Up to 275 GeV:** fewer Λs decay before the B0 (26 % at 275 GeV), but they are reconstructed better $\Rightarrow$ fully reconstructed fraction roughly flat (old 15–25 %, new 9–14 %)
- Use **per generated** Λ: "per $p\pi^-$ decay" is conditional on an early decay

Open: old-geometry efficiency jump between EICrecon builds; CKF loss of soft tracks; +2 MeV mass shift (old geometry); follow #3116 until merged; full events with backgrounds and vertexing

Backup: reproducing

- `scripts/build/build_eicrecon.sh EICrecon_pr3116 3116;`
  `scripts/build/build_epic.sh epic_43f427e <commit>`
- `scripts/production/lambda_rereco.sh <GEO> <NAME> EICrecon_pr3116 eicreconpr3116`
- `PMIN=100 PMAX=275 SIM_ONLY=1 scripts/production/lambda_sample.sh ...`
- `macros/efficiency/lambda_vs_momentum.C`,
  `macros/mass/lambda_mass_seeding_compare.C`
- `scripts/make_results.sh 07 08`

Pitfalls

- The nightly changes under you: `GEO=container` is no longer the old geometry ⇒ pin commits
- JANA keeps the *first* value of a repeated `-P` (`jana:nevents`)